CareSphere-AI Web Scraping Tool for Care Assistance: A Comprehensive Review

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Abstract—Ageing in place increases demand for digital systems that go beyond health monitoring to connect older adults with caregivers, support communication and provide timely assistance. This paper reviews fourteen studies on elder–caregiver coordination, smart-home and context-aware care, cloud/digital-twin healthcare, conversational assistance, caregiver allocation and service governance, drawn from IEEE, Springer/IFIP, SHS Web of Conferences and similar venues. The review shows that existing solutions address isolated functions and rarely combine caregiver discovery, matching, booking, coordination, safety, payments and administration. Based on these findings, we propose the CareSphere Caregiver-Centered Integrated Eldercare Platform (CCIEP) and prototype its caregiver-matching module as a two-stage pipeline: deterministic eligibility filtering followed by a Random Forest relevance ranker (care-ranker-v1) with a rule-based fallback. On a held-out synthetic test set of 300 queries with eight candidates each, the ranker reached an NDCG@5 of 0.3956 against 0.3599 for a weighted rule baseline (+9.9% relative) and a ROC-AUC of 0.6895 against 0.6501. These results are offline, rely on rule-generated labels and do not establish real-world care quality; we therefore report limitations and a validation plan based on real feedback.

Keywords—*eldercare, home healthcare, caregiver coordination, smart home, Internet of Things, artificial intelligence, caregiver allocation, caregiver recommendation, learning to rank, random forest, digital health, service ecosystems, ageing in place*

I. INTRODUCTION

Digital eldercare research has evolved in several distinct but related directions. Early work focused on enabling computer-assisted coordination among the human caregivers involved in a home-based elderly care service. Subsequent work has focused on introducing smart-home sensing, context-aware reasoning, telehealth, cloud healthcare, artificial intelligence, and service ecosystems. More recent work has also focused on algorithmic approaches to caregiver allocation and the assessment of digital-home-care quality. The most relevant contributions for CareSphere, therefore, are the intersection of these research areas: an older adult should be able to access an appropriate care service provided by an assigned caregiver, while the system coordinates the care relationship, information, safety, and operational workflow.

Within such a platform, deciding which caregiver to show an elder first is a ranking problem under hard constraints: a caregiver who is unverified, unavailable, over budget or too far away must never be recommended, whereas a learned score can order the remaining eligible candidates. This paper makes three contributions:

1) a structured review of fourteen studies on digital eldercare, with a comparative analysis and a research gap centered on caregiver-service integration;
2) the CareSphere Caregiver-Centered Integrated Eldercare Platform (CCIEP), a platform-level synthesis of the reviewed work; and
3) a prototype of the CCIEP caregiver-matching module (care-ranker-v1) in which a Random Forest ranker orders candidates that have already passed deterministic eligibility checks, with a transparent offline evaluation against a rule-based baseline.

Section II states the review scope, Section III reviews the models, Section IV compares them and Section V identifies the research gap. Section VI proposes CCIEP, Section VII maps it to the literature, Section VIII presents the machine-learning ranking subsystem and its offline evaluation, Section IX outlines the evaluation plan and Section X concludes.

II. REVIEW SCOPE AND PRIORITY

The review covers fourteen studies selected for their relevance to connecting older adults with caregivers and home-care services. Only studies whose publication metadata could be verified directly against the publisher or repository record are cited as formal references, which avoids unverifiable author, venue or DOI information. The studies are grouped into three priority levels:

- Priority 1 — direct elder–caregiver/service integration: CSCC, Care@Home, CARA, smart elderly-care service models and caregiver-allocation research.

- Priority 2 — digital home-care and service ecosystem research: digital home care, service-level agreements and smart-home goal models.
- Priority 3 — enabling healthcare technologies: cloud/digital twin, smartphone safety and socially assistive robotics.

III. DETAILED REVIEW OF MODELS

A. *Smart Elderly Care Services in Home Elderly Care*

Shi and Zhou [1] presented a smart elderly-care service model that integrates artificial intelligence, Internet of Things technologies, and big-data analytics to support personalized elderly care. The model addresses health monitoring, assistance with daily activities, and emergency response within a home-based care environment. The approach also illustrates how digital technologies can complement responsibilities traditionally performed by family caregivers.

However, the study is primarily conceptual and relies on case-based discussion rather than a fully implemented caregiver-service system. The authors identify challenges associated with interoperability, data accuracy, privacy, security, cost, and long-term sustainability. In addition, the model does not provide experimental validation of a complete caregiver booking and service-delivery workflow. [1]

B. *Technology for Care Networks of Elders*

Consolvo et al. [2] proposed Computer-Supported Coordinated Care (CSCC), an approach designed to use technology to support groups of people involved in caring for an older adult at home. The model emphasizes communication, coordination, and distribution of caregiving responsibilities to enable older adults to remain at home as their care requirements increase.

Although the CSCC model establishes an important foundation for technology-supported care coordination, it predates current cloud, smartphone, and digital marketplace architectures. Consequently, it does not address modern caregiver discovery, automated matching, service booking, digital payments, escrow mechanisms, or AI-based assistance. Its evaluation also does not represent the large-scale deployment characteristics expected from contemporary digital care platforms. [2]

C. *A SmartTV Platform for Wellbeing, Care and Social Support for Elderly at Home*

Fitrianie et al. [3] introduced Care@Home, a SmartTV-based platform designed to connect older adults with formal care networks, family members, friends, and community services. The platform supports multiple aspects of elderly well-being, including household assistance, healthcare, exercise, and social or entertainment activities.

The platform demonstrates the potential of user-centered technology to connect older adults with different sources of support. However, its interaction model is strongly dependent on SmartTV technology, which limits portability compared with contemporary mobile and web-based applications. Furthermore, the reported work focuses on human-centered design and early user studies rather than long-term, large-scale validation of caregiver scheduling, payment, dispute management, and care outcomes. [3]

D. *Hybrid Reasoning Framework for CARA Pervasive Healthcare*

Yuan and Herbert [4] proposed CARA, a Context Aware Real-time Assistant, for providing personalized and timely healthcare services to older adults. The framework adapts technological assistance to normal activities and caregiver practices by combining context-aware sensor-data fusion, activity-of-daily-living analysis, anomaly detection, rule-based reasoning, and case-based reasoning.

CARA demonstrates the usefulness of contextual information and intelligent reasoning in elderly care. Nevertheless, its primary focus is smart-home sensing and healthcare assistance rather than caregiver-service coordination. The framework does not address caregiver discovery, availability-based matching, service booking, payment processing, dispute management, or complete administrative supervision. [4].

E. *Elderly Safety: A Smartphone Based Real Time Approach*

Alam et al. [5] developed a smartphone-based approach for real-time elderly safety and activity recognition. The approach aims to reduce dependence on external sensor installations by using smartphone-based information for recognizing activities and supporting elderly safety.

The approach is primarily concerned with safety and activity recognition rather than caregiver-service management. Consequently, a smartphone-centered safety mechanism alone does not provide caregiver selection, booking, care-plan management, payment facilities, or administrative supervision. These limitations indicate the need to combine safety technologies with broader caregiver coordination mechanisms. [5]

F. *Development of the Smart Home System for Elder Healthcare Based on IoT Technology*

Carol and Hsieh [6] developed an IoT-based smart-home system for elder healthcare. The system supports healthcare monitoring and smart-home assistance, demonstrating how information generated by IoT devices can potentially support remote care and intervention.

Although the system is relevant to digital eldercare, it represents a smart-home healthcare system rather than a complete caregiver marketplace. The available study information does not establish long-term clinical deployment or broad validation of caregiver workflows. Functions such as caregiver booking, payment, and service governance are outside the primary scope of the system. [6]

G. GER-CARE, an Elderly Assistant Chatbot

Deshmukh et al. [7] presented GER-CARE, a conversational assistant designed to support elderly users. A conversational interface can provide reminders, information, and assistance while providing an accessible interaction mechanism for older adults. This functionality is particularly relevant to the AI component of an integrated eldercare platform.

However, GER-CARE represents an assistance component rather than a complete caregiver-management platform. The available publication information does not demonstrate comprehensive real-world evaluation of caregiver allocation, booking, escrow-based payments, or care-quality measurement. Therefore, conversational assistance can be considered one component of a broader eldercare ecosystem rather than a complete solution. [7]

H. A Novel Cloud-Based Framework for the Elderly Healthcare Services Using Digital Twin

Liu et al. [8] proposed CloudDTH, a cloud-based Digital Twin Healthcare framework that integrates cloud computing, IoT, and digital-twin technologies for monitoring, diagnosing, and predicting health conditions in elderly users. The framework implements a digital-twin healthcare model and demonstrates application scenarios involving real-time supervision.

The framework provides capabilities for healthcare-data management and monitoring but remains primarily healthcare-centric rather than caregiver-service-centric. It does not provide an end-to-end mechanism for connecting older adults with caregivers, managing service bookings, processing payments or escrow transactions, or handling caregiver-related disputes. [8]

I. Enhancing Elderly Care With Socially Assistive Robots: A Holistic Framework for Mobility, Interaction, and Well-Being

Naseer et al. [9] presented a holistic framework for socially assistive robots that addresses mobility assistance, social interaction, and well-being support for older adults. The framework demonstrates how technological systems can complement human caregiving while addressing both physical and psychosocial aspects of elderly care.

Despite these capabilities, the approach depends on specialized robotic hardware and focuses primarily on assistive interaction rather than human caregiver marketplace and service coordination. Hardware costs, user acceptance, cultural adaptability, and deployment complexity may also influence practical scalability. [9]

J. Decision Support Framework for Home Health Caregiver Allocation Using Optimally Tuned Spectral Clustering and Genetic Algorithm

Sharifnia et al. [10] proposed a decision-support framework for home-health caregiver allocation using optimally tuned spectral clustering and a genetic algorithm. The approach groups patients and allocates caregivers while considering travel mileage, flexible visit sequences, workload, and continuity of care. In a Tennessee home-health agency case study, the authors reported reductions of up to 42% in average travel mileage depending on the discipline.

This work is particularly relevant to the caregiver-matching component of CareSphere because it demonstrates how optimization techniques can improve caregiver allocation. However, the framework focuses on operational allocation rather than the complete patient-facing service lifecycle. The study also indicates that practical implementation may produce smaller improvements than algorithmic case-study results because organizational structures, clinician preferences, and communication can influence adoption. [10]

K. Managing Performance in Technology-Enabled Elderly Care Services: The Role of Service Level Agreements in Modular Smart Service Ecosystems

Mugurusi et al. [11] examined technology-enabled elderly-care services through a modular smart-service ecosystem and Service-Level Agreement (SLA) perspective. The study discusses case management, integrated care, consumer-directed care, and restorative care models and examines how SLAs can help coordinate performance among different actors.

This perspective is valuable for defining responsibilities and service expectations within a CareSphere ecosystem. Nevertheless, the study is primarily organizational rather than a fully implemented software platform. It also identifies trade-offs associated with different care models, including resource requirements, coordination challenges, institutional dependence, and shifts in cost and control toward consumers. [11]

L. Digital Home Care Interventions and Quality of Primary Care for Older Adults: A Scoping Review

Silva et al. [12] conducted a scoping review of digital home-care interventions for older adults. The reviewed interventions include tele monitoring, telehealth, mobile applications, sensors, video communication, and other digital platforms. The review highlights access to home caregivers and the involvement of patients and caregivers in health-related decisions and technology design.

The study also identifies challenges related to digital literacy, infrastructure, economic conditions, communication, and equity. As a scoping review rather than a single platform implementation, it does not define a complete caregiver-management architecture. Furthermore, some reviewed interventions did not demonstrate clear improvements in quality-of-life or health outcomes. [12]

M. Smart Home Goal Feature Model - A Guide to Support Smart Homes for Ageing in Place

Logothetis et al. [13] proposed the Smart Home Goal Feature Model (SHGFM) as a design-oriented approach for

supporting ageing in place. The model structures goals and features for personalized smart homes and includes safety, health and environmental monitoring, emergency detection, and notification of informal and formal caregivers.

SHGFM provides useful design guidance for integrating elderly-care requirements into smart-home systems. However, it is not a complete deployed caregiver-service platform. It does not define a full caregiver marketplace, service transaction mechanism, payment process, care-session workflow, or large-scale empirical validation. [13]

N. Research on the Development and Countermeasures of Smart Elderly Care Demand in Chongqing

Zhao et al. [14] investigated smart elderly-care demand and acceptance in Chongqing and discussed service models such as community comprehensive elderly-service centre and Internet-based one-stop intelligent elderly-care platforms. These platforms may provide services including appointment management, policy consultation, and elderly-care consultation.

The study is primarily concerned with demand analysis, policy, and service models rather than technical implementation of caregiver management. It identifies challenges related to awareness, usability, privacy, security, and service adoption but does not evaluate an integrated caregiver-booking system. [14]

IV. COMPARATIVE ANALYSIS

Table I compares ten of the reviewed models by caregiver priority, main function, platform type, key limitation and intended use within CareSphere.

TABLE I
COMPARATIVE ANALYSIS OF REVIEWED MODELS

Paper / Model	Priority	Main Function	Platform Type	Key Limitation	CareSphere Use
CSCC [2]	Very High	Care-network coordination	Coordination platform	Old technology; no transactions	Caregiver communication + coordination
Care@Home [3]	Very High	Care, social and household services	Service hub	SmartTV-centric	Elder-facing service hub
CARA [4]	High	Context-aware alerts and personalized assistance	Smart-home assistant	Sensor/logic dependent	Safety + caregiver alerts
Smart Elderly Care (Shi & Zhou) [1]	High	Personalized smart home/community care	Service model	Conceptual; privacy/cost issues	Overall service ecosystem
Caregiver Allocation Framework [10]	Very High	Patient–caregiver allocation	Optimization/DSS	Allocation only; adoption constraints	Matching + scheduling
APMS Smart Service Ecosystem [11]	Very High	Service governance and actor coordination	Service ecosystem	Organizational rather than software implementation	Roles, SLAs and care governance
CloudDTH [8]	Medium	Health monitoring/prediction	Cloud digital twin	Not caregiver-workflow centric	Health data layer
GER-CARE [7]	Medium	Conversational assistance	AI chatbot	Not complete care platform	AI assistant
Smart Home Goal Feature Model [13]	High	Goal/feature design	Design framework	No full transaction workflow	Requirements/architecture
Digital Home Care Scoping Review [12]	Very High	Evidence across digital home care	Evidence framework	Review, not platform	Design requirements + evaluation

V. RESEARCH GAP

While the literature offers viable individual solutions, the solutions are rarely integrated. CSCC articulates the value of a coordinated care network; Care@Home embodies a user-facing service hub; CARA exemplifies context-aware assistance; the caregiver-allocation framework provides a quantitative matching mechanism; APMS realizes service-level governance; CloudDTH demonstrates cloud health-data integration; and recent smart-care studies demonstrate AI/IoT-enabled personalized care. However, the functions are typically distributed across disparate systems. A caregiver-first platform can offer to bridge a number of these gaps.

- Caregiver discovery and selection is not always consistently integrated with health/service coordination.
- Caregiver matching is often viewed as an optimization problem rather than as a complete elder-facing workflow.
- Booking, acceptance/rejection, completion, rescheduling and service history are rarely combined with smart-health functions.
- Payment, escrow, disputes and administrative accountability are largely absent from the reviewed eldercare models.
- AI assistance is usually realized as a separate component rather than part of a caregiver-centered service ecosystem.
- Privacy, digital literacy, interoperability, infrastructure and cost remain recurring adoption barriers.
- Long-term, multi-site, real-world evaluation is limited across many prototypes and conceptual models.

- Among the reviewed models, none evaluates a learned ranker for elder-facing caregiver recommendation against an explicit rule-based baseline.

VI. PROPOSED CARESPHERE MODEL

Proposed model name: CareSphere Caregiver-Centered Integrated Eldercare Platform (CCIEP)

We propose that the model represent a platform-level synthesis of the best ideas identified in the reviewed papers, with the caregiver relationship as the highest-priority function. The platform is not a model taken from any single reviewed paper; rather, it is the proposed architecture for the CareSphere project.

1) Elder profile and care-needs assessment - capture preferred services, schedule, location, emergency contacts and care requirements.
2) Caregiver marketplace - verified caregiver profiles with skills, experience, rating, rate, availability and service area.
3) Intelligent caregiver matching - combine required skills, availability, distance, rating, rate and continuity-of-care preferences; the caregiver-allocation literature provides the optimization foundation. In the prototype this component is a two-stage eligibility filter and learned ranker (Section VIII).
4) Booking and care workflow - request, accept/reject, schedule, reschedule, start, complete and review a care session.
5) Care coordination - care instructions, messages, notifications, caregiver status updates and escalation.
6) Safety and health-support layer - optional IoT/health-monitoring integration inspired by CARA, smart-home and CloudDTH research.
7) AI elder assistant - multilingual conversational support, reminders and navigation of platform services; not a substitute for a clinician.
8) Healthcare/facility discovery - hospitals, clinics, pharmacies and other facilities can be surfaced to support the elder/caregiver journey.
9) Trust, payment and governance - escrow, completion confirmation, dispute management, role-based access and audit logs.
10) Admin and quality layer - caregiver verification, service monitoring, analytics, complaints and system-level performance indicators.



Fig. 1. Proposed workflow of the CareSphere Caregiver-Centered Integrated Eldercare Platform (CCIEP), from elder profiling through caregiver matching, booking, coordination, safety, AI assistance, facility discovery, trust/payment and administrative quality control. The matching step is detailed in Fig. 2.

VII. PROPOSED ARCHITECTURE MAPPING TO LITERATURE

TABLE II
MAPPING OF CARESPHERE COMPONENTS TO LITERATURE

CareSphere Component	Research Basis	How CareSphere Extends It
Caregiver network	CSCC	Turns coordination into a persistent digital caregiver–elder service relationship.
Service hub	Care@Home	Moves beyond SmartTV to web/mobile platform and transactional care workflows.
Context-aware safety	CARA	Adds caregiver escalation and service actions after alerts.
Caregiver matching	Sharifnia et al.	Adds elder-facing search, preferences, availability and booking to optimization.
Service governance	Mugurusi et al.	Adds operational roles, audit trails, disputes and measurable service performance.
Smart care	Shi & Zhou	Combines AI/IoT concepts with actual caregiver service delivery.
Digital health	CloudDTH	Provides a future-ready health-data integration layer.
AI assistant	GER-CARE	Embeds conversational assistance inside the caregiver workflow.
Ageing-in-place requirements	SHGFM	Maps personalized goals to concrete platform features.

VIII. MACHINE-LEARNING CAREGIVER RANKING SUBSYSTEM

To make the matching function of CCIEP concrete, its caregiver-matching module was prototyped within the TrustCare Management System [15], a web application with a Next.js frontend, an Express backend, a FastAPI inference service and a MongoDB Atlas database, deployed on Vercel and Render. This section documents the ranking subsystem and its offline evaluation. It does not address diagnosis, treatment effectiveness or patient outcomes.

A. Two-Stage Ranking Architecture

The module separates policy from statistics (Fig. 2). In Stage 1, the Express backend applies deterministic eligibility constraints: a candidate must be verified, available, offer the requested care type, satisfy the budget and lie within 50 km. Only eligible candidates reach Stage 2, in which a FastAPI [16] service loads the serialized model care-ranker-v1 and returns, for each candidate, the positive-class probability of a Random Forest classifier [17] as a relevance score. The backend sorts candidates by descending score and serves caregiver and facility recommendations through separate API endpoints. If the service or model is unavailable, the backend falls back to a weighted rule score (0.30 care-type match, 0.25 distance, 0.25 rating, 0.10 experience and 0.10 coverage) multiplied by verified and availability indicators. The score is used only for ordering. It is neither a calibrated probability nor a clinical risk estimate, and the model never overrides the eligibility rules.

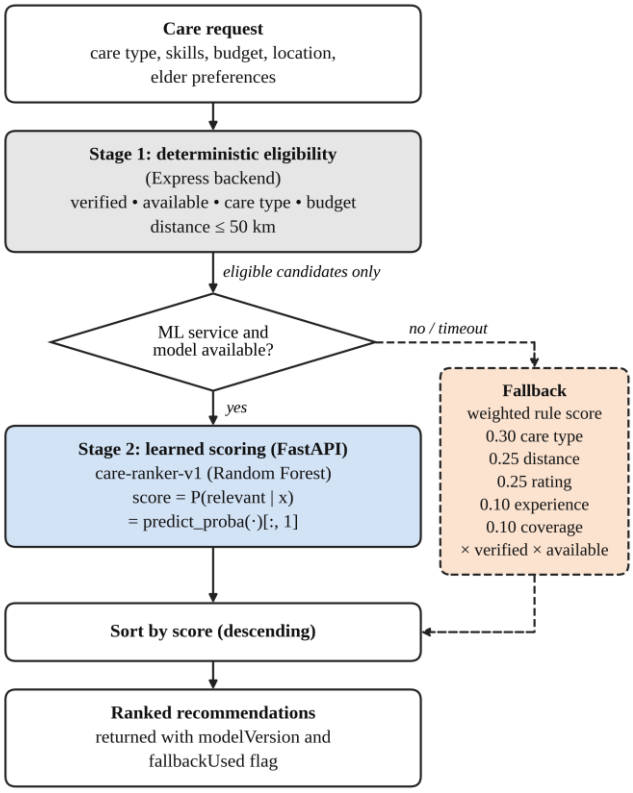


Fig. 2. Two-stage caregiver ranking pipeline of the CCIEP matching module. Deterministic eligibility filters precede learned scoring by care-ranker-v1; a weighted rule score (dashed path) is used when the ML service or model is unavailable.

B. Dataset and Label Construction

care-ranker-v1 was trained and evaluated on a synthetic dataset of 12,000 request–candidate pairs (1,500 requests × 8 candidates, random seed 42) covering 8 care types, 10 skills and 7 Indian cities (Fig. 3). Binary relevance labels follow a documented synthetic rule with 8% label noise, giving a positive prevalence of 11.5%. The labels therefore encode the assumptions of that rule rather than observed user satisfaction, caregiver quality or patient outcomes. Separately, the live prototype holds 50 facility records gathered from OpenStreetMap [18], Nominatim [19] and Wikidata [20] and 50 caregiver records in MongoDB; these operational records are not part of the offline evaluation.

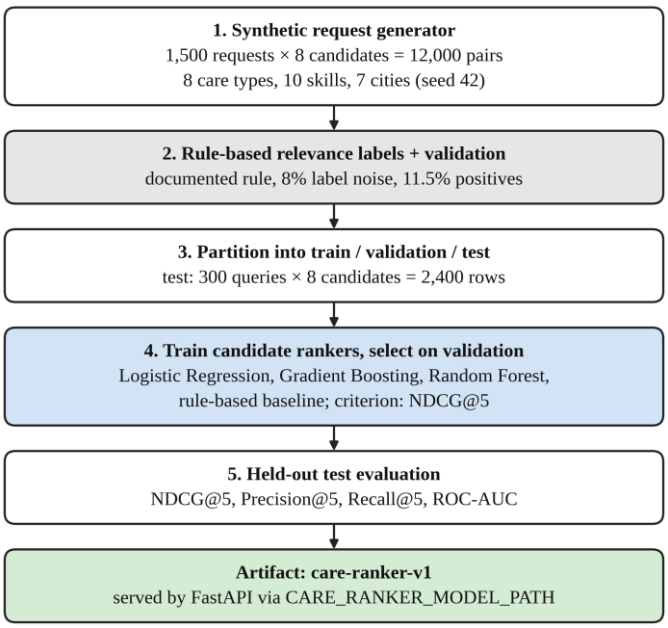


Fig. 3. Offline experimental protocol for care-ranker-v1: synthetic data generation, rule-based labelling, partitioning, validation-based model selection and held-out test evaluation.

C. Model and Training Configuration

The selected model is a scikit-learn [21] RandomForestClassifier used as a pointwise ranker: candidates are ordered by predict_proba(x)[:, 1]. Table III lists the configuration. Logistic Regression, Gradient Boosting and a rule-based baseline were compared with the Random Forest, which was selected by NDCG@5 on a held-out validation split; test scores of the two other learned models are not reported in this version. The pipeline is run with build_dataset.py, validate_data.py, train.py, evaluate.py and a pytest suite, and the exact feature schema, encodings and label rule are given in the repository [15].

TABLE III
DATASET AND MODEL CONFIGURATION

Item	Value
Dataset	12,000 pairs (1,500 requests × 8 candidates)
Coverage	8 care types, 10 skills, 7 Indian cities
Label noise / positives	8% / 11.5%
Test partition	300 queries, 2,400 rows (88.3% negative)
Estimator	RandomForestClassifier (scikit-learn 1.5.2)
n_estimators / max_depth	300 / 12
min_samples_leaf / class_weight	5 / balanced
Random seed (data, model)	42, 42

D. Evaluation Protocol and Metrics

Models are evaluated per query on the held-out test partition of 300 queries (2,400 rows), each with eight candidates. We report NDCG@5 [22], which rewards placing relevant candidates near the top, Precision@5 and Recall@5 [23] averaged over queries, and ROC-AUC for pairwise discrimination. Because 88.3% of test pairs are negative, we also give threshold-0.5 confusion counts but do not treat accuracy as a headline metric [24], [25].

E. Results and Discussion

Table IV and Fig. 4 summarize the test results. Relative to the rule baseline, the Random Forest raised NDCG@5 from 0.3599 to 0.3956 (+0.0357, +9.9%) and ROC-AUC from 0.6501 to 0.6895 (+0.0394, +6.1%), whereas Precision@5 (0.1420 to 0.1447) and Recall@5 (0.4791 to 0.4949) improved only marginally. The test partition contains 281 relevant pairs, about 0.94 per query, and a random ordering would give a Precision@5 near the positive rate of about 0.117; absolute Precision@5 values are therefore necessarily low in this sparse-positive setting.

TABLE IV
HELD-OUT TEST RESULTS (300 QUERIES, 2,400 ROWS)

Metric	Rule baseline	Random Forest	Abs. Δ	Rel. Δ
Precision@5	0.1420	0.1447	+0.0027	+1.9%
Recall@5	0.4791	0.4949	+0.0158	+3.3%
NDCG@5	0.3599	0.3956	+0.0357	+9.9%
ROC-AUC	0.6501	0.6895	+0.0394	+6.1%

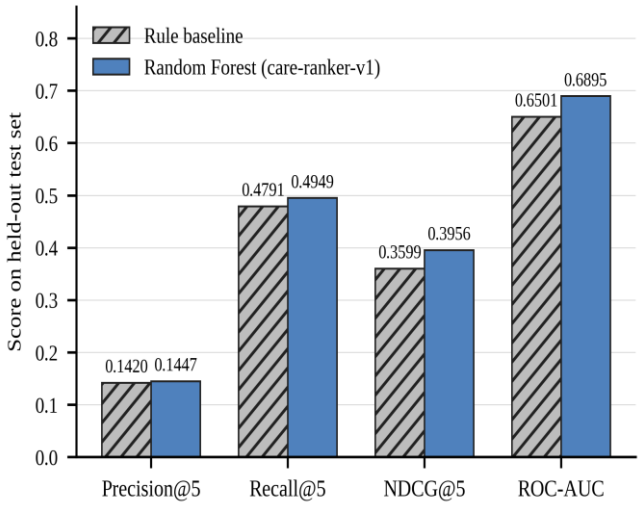


Fig. 4. Held-out test performance of the rule baseline and care-ranker-v1 (Random Forest).

At a 0.5 threshold the confusion counts are TP = 108, FP = 121, FN = 173 and TN = 1,998, giving accuracy 0.8775, precision 0.4716, recall 0.3843 and F1 0.4235. A classifier that labels every pair negative would reach an accuracy of 0.8829 on this test set, which is higher, so accuracy is not used to judge the ranker. In a local end-to-end check the service returned modelVersion care-ranker-v1 with fallbackUsed false, the constraint-checking path recorded zero eligibility violations in the tested cases, and scoring took about 0.03 ms per row.

F. Limitations and Threats to Validity

These findings are preliminary. (i) Labels come from a hand-written synthetic rule, so the ranker partly learns that rule; the results do not demonstrate real-world relevance, user preference or care outcomes. (ii) Rows from one request are dependent; request-level disjointness of the partitions, confidence intervals and multi-seed repeats are still to be reported. (iii) Pointwise classification ignores list structure, and pairwise or listwise objectives such as RankNet [26] remain to be compared. (iv) Scores are not calibrated, no subgroup analysis by city, provider, language, disability or socioeconomic status has been performed, and the latency measurement conditions are not yet documented. (v) Source licensing, freshness and consent for operational records must be verified before deployment. The ranker orders eligible candidates only and must not be presented as medical advice.

IX. EVALUATION PLAN FOR THE PROPOSED PLATFORM

Ranking quality has been assessed offline (Section VIII); the remaining indicators require a pilot deployment with real elders, families and caregivers.

- Caregiver matching accuracy: percentage of recommended carers matching the required skills and availability.
- Booking completion rate: accepted bookings which reach completion.

- Response time: interval between elder's request and carer's acceptance.
- Continuity of care: percentage of repeat visits provided by the same caregiver if requested.
- Travel efficiency: average carer travel distance per completed visit.
- Elder usability: task-completion rate and usability score for booking and communication.
- Caregiver usability: time needed to update availability and manage bookings.
- Safety response: interval between alert trigger and caregiver/admin acknowledgement.
- Trust and security: auditability of sensitive operations and successful access-control tests.
- Service quality: elder satisfaction, caregiver satisfaction, cancellation rate and dispute rate.
- Ranking quality: NDCG@5, Precision@5, Recall@5 and ROC-AUC of the learned ranker against the rule baseline, using request-grouped held-out data and bootstrap confidence intervals.
- Fairness and coverage: ranking metrics stratified by city, provider type and care type, with checks for systematic exposure gaps.

X. CONCLUSION

The reviewed research provides a consistent direction for CareSphere. The most compelling caregiver-focused foundations are represented by CSCC, Care@Home, CARA, the smart elderly-care service model, the caregiver-allocation framework and the modular smart-service ecosystem. Altogether, this body of work demonstrates that effective eldercare extends beyond sensing and AI: it requires coordination among the elder, caregiver, family/community, healthcare resources and the service operator. The proposed CCIEP model integrates these features in a single platform, with caregiver discovery, matching and service delivery as the highest-priority layer.

As a first step toward realizing CCIEP, the caregiver-matching module was prototyped as a two-stage pipeline in which deterministic eligibility checks precede a Random Forest relevance ranker, with a weighted rule score as fallback. On held-out synthetic data the ranker improved NDCG@5 from 0.3599 to 0.3956 and ROC-AUC from 0.6501 to 0.6895 over the rule baseline. Because the labels are rule-generated, the gains are modest and uncertainty estimates are not yet available, these results demonstrate a reproducible offline ranking prototype rather than real-world care quality. Future work will (i) use request-grouped cross-validation with confidence intervals across seeds, (ii) report all candidate models, (iii) replace synthetic labels with real feedback from elders, families and caregivers, (iv) audit ranking fairness across cities and provider types, and (v) evaluate the full platform in a pilot deployment.

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